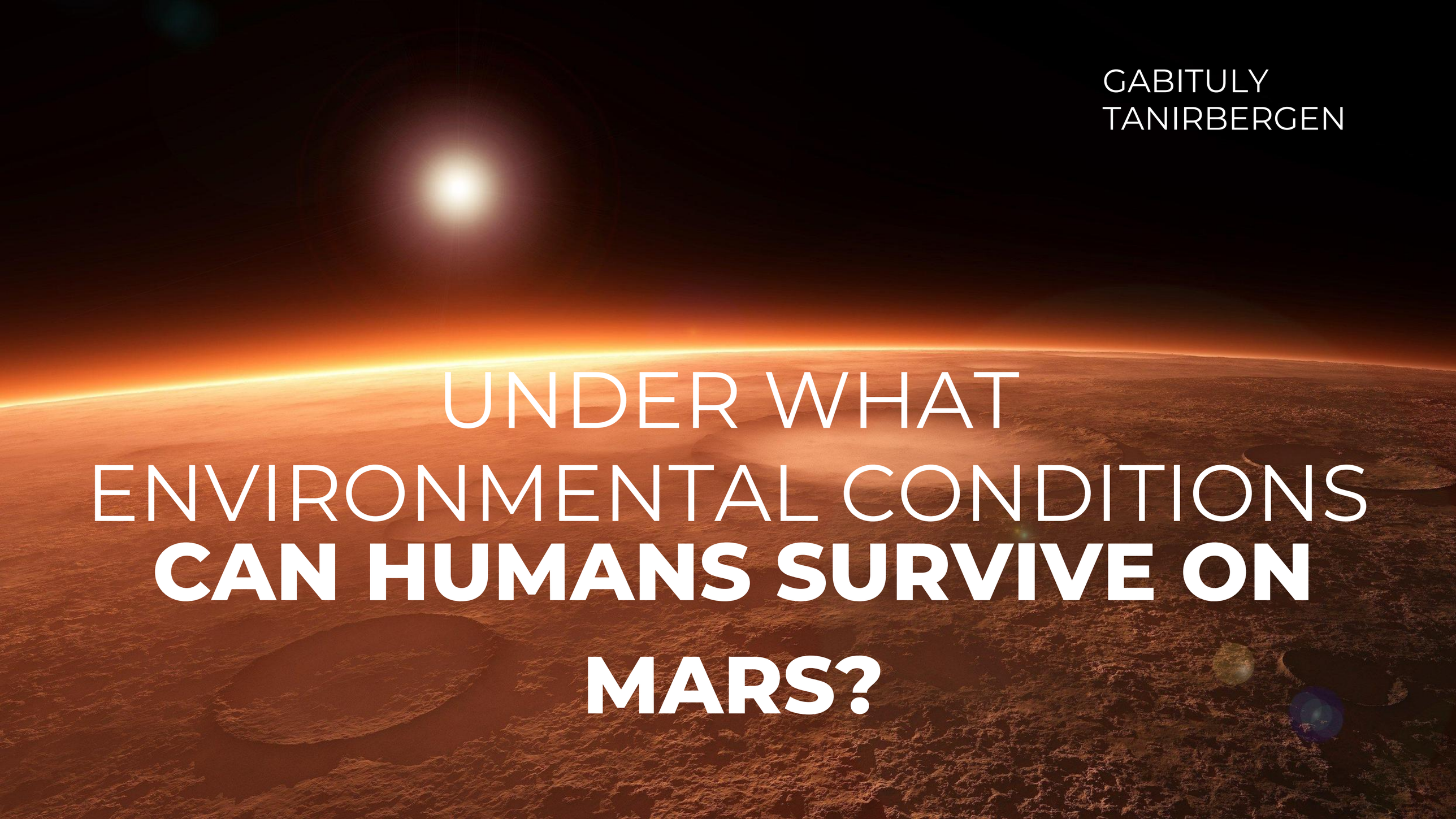

GABITULY
TANIRBERGEN
UNDER WHAT
ENVIRONMENTAL CONDITIONS
CAN HUMANS SURVIVE ON
MARS?

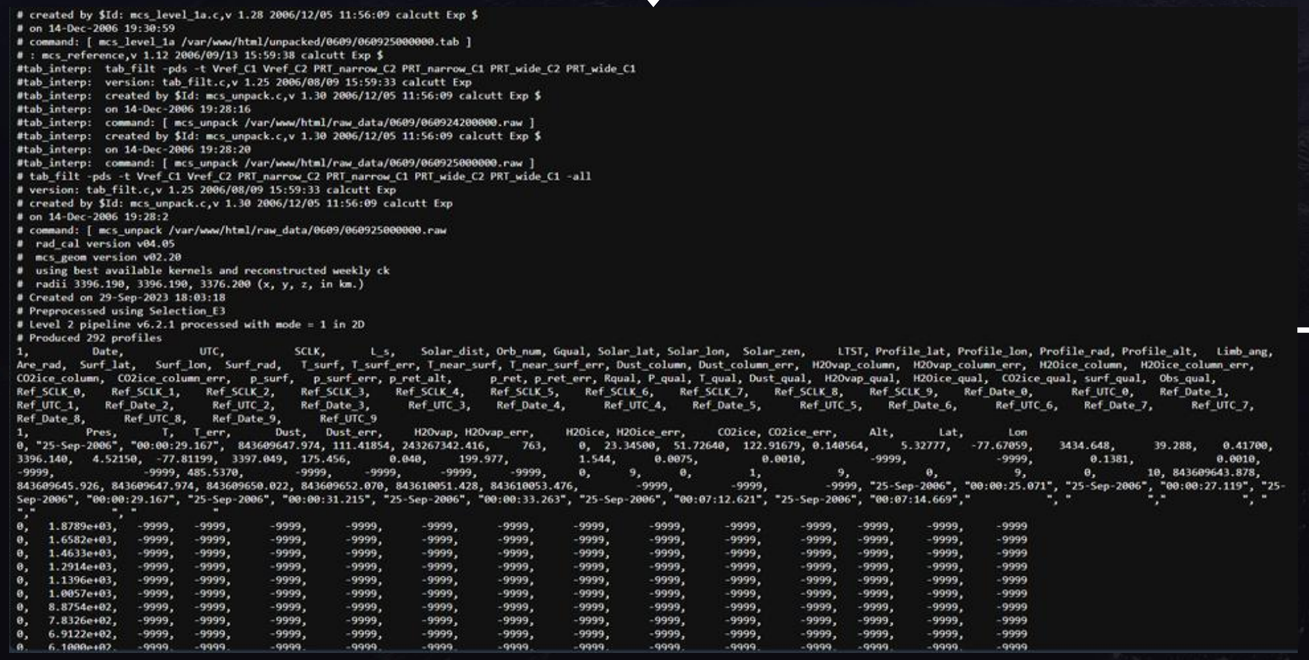
AGENDA

1. DATA COLLECTION
2. INTRODUCTION
3. DATA CLEANING & PREPARATION
4. EXPLORATORY DATA ANALYSIS (EDA)
5. TIME SERIES ANALYSIS (ARIMA)
6. HYPOTHESIS TESTING
7. CONCLUSION



DATA COLLECTION

Collection labels and inventory tables usually reside in the top-level subdirectories. If there are many products in the collection they are usually divided into lower level subdirectories as shown in Figure 2B-1. The lower level subdirectories are named and organized at the discretion of the data provider except for SPICE kernel subdirectories, which have a required structure (see Section 2B.2.2.3). An example directory structure may be found on the PDS4 Examples web page at https://pds.nasa.gov/datastandards/documents/examples/.



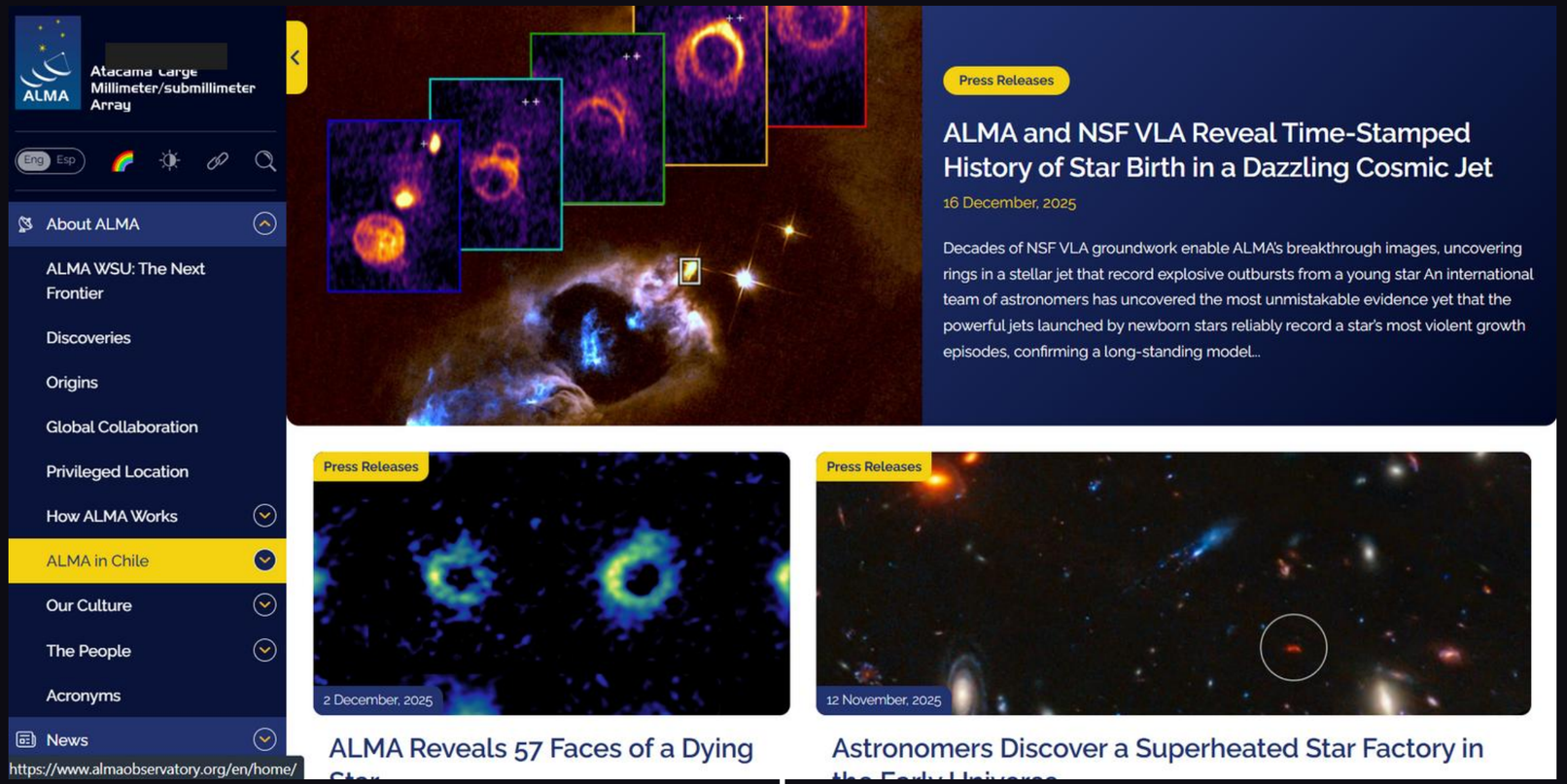
```
# created by $Id: mcs_level_1a.c,v 1.28 2006/12/05 11:56:09 calcutt Exp $
# on 14-Dec-2006 19:30:59
# command: [ mcs_level_1a /var/www/html/unpacked/0609/060925000000.tab ]
# : mcs_reference,v 1.12 2006/09/13 15:59:38 calcutt Exp $
#tab_interp:  tab_filt -pds -t Vref_C1 Vref_C2 PRT_narrow_C2 PRT_narrow_C1 PRT_wide_C2 PRT_wide_C1
#tab_interp:  version: tab_filt.c,v 1.25 2006/08/09 15:59:33 calcutt Exp
#tab_interp:  created by $Id: mcs_unpack.c,v 1.30 2006/12/05 11:56:09 calcutt Exp $
#tab_interp:  on 14-Dec-2006 19:28:16
#tab_interp:  command: [ mcs_unpack /var/www/html/raw_data/0609/060924200000.raw ]
#tab_interp:  created by $Id: mcs_unpack.c,v 1.30 2006/12/05 11:56:09 calcutt Exp $
#tab_interp:  on 14-Dec-2006 19:28:20
#tab_interp:  command: [ mcs_unpack /var/www/html/raw_data/0609/060925000000.raw ]
# tab_filt -pds -t Vref_C1 Vref_C2 PRT_narrow_C2 PRT_narrow_C1 PRT_wide_C2 PRT_wide_C1 -all
# version: tab_filt.c,v 1.25 2006/08/09 15:59:33 calcutt Exp
# created by $Id: mcs_unpack.c,v 1.30 2006/12/05 11:56:09 calcutt Exp
# on 14-Dec-2006 19:28:2
# command: [ mcs_unpack /var/www/html/raw_data/0609/060925000000.raw
#  rad_cal version v04.05
#  mcs_geom version v02.20
#  using best available kernels and reconstructed weekly ck
#  radii 3396.190, 3396.190, 3376.200 (x, y, z, in km.)
# Created on 29-Sep-2023 18:03:18
# Preprocessed using Selection_E3
# Level 2 pipeline v6.2.1 processed with mode = 1 in 2D
# Produced 292 profiles
1,          Date,          UTC,          SCLK,       L_s,    Solar_dist, Orb_num, Gqual, Solar_lat, Solar_lon,  Solar_zen,     LTST, Profile_lat, Profile_lon, Profile_rad, Profile_alt,   Limb_ang,
Are_rad,  Surf_lat,   Surf_lon,  Surf_rad,  T_surf, T_surf_err, T_near_surf, T_near_surf_err, Dust_column, Dust_column_err,  H2Ovap_column,  H2Ovap_column_err,  H2Oice_column,  H2Oice_column_err,
CO2ice_column,  CO2ice_column_err,   p_surf,   p_surf_err, p_ret_alt,     p_ret, p_ret_err, Rqual, P_qual, T_qual, Dust_qual,  H2Ovap_qual,  H2Oice_qual,  CO2ice_qual, surf_qual,  Obs_qual,
Ref_SCLK_0,    Ref_SCLK_1,    Ref_SCLK_2,    Ref_SCLK_3,    Ref_SCLK_4,    Ref_SCLK_5,    Ref_SCLK_6,    Ref_SCLK_7,    Ref_SCLK_8,    Ref_SCLK_9,    Ref_Date_0,      Ref_UTC_0,    Ref_Date_1,
Ref_UTC_1,    Ref_Date_2,      Ref_UTC_2,    Ref_Date_3,      Ref_UTC_3,    Ref_Date_4,      Ref_UTC_4,    Ref_Date_5,      Ref_UTC_5,    Ref_Date_6,      Ref_UTC_6,    Ref_Date_7,      Ref_UTC_7,
Ref_Date_8,      Ref_UTC_8,    Ref_Date_9,      Ref_UTC_9
1,          Pres,          T,   T_err,       Dust,   Dust_err,     H2Ovap, H2Ovap_err,     H2Oice, H2Oice_err,     CO2ice, CO2ice_err,     Alt,       Lat,       Lon
0, "25-Sep-2006", "00:00:29.167",  843609647.974, 111.41854, 243267342.416,     763,     0,  23.34500,  51.72640,  122.91679, 0.140564,     5.32777,   -77.67059,    3434.648,      39.288,    0.41700,
3396.140,   4.52150,  -77.81199,  3397.049,  175.456,      0.040,     199.977,      1.544,      0.0075,          0.0010,          -9999,          -9999,          0.1381,          0.0010,
-9999,          -9999, 485.5370,          -9999,     -9999,     -9999,     -9999,     0,     9,     0,     1,     9,     0,     9,     0,     10, 843609643.878,
843609645.926, 843609647.974, 843609650.022, 843609652.070, 843610051.428, 843610053.476,          -9999,          -9999,          -9999, "25-Sep-2006", "00:00:25.071", "25-Sep-2006", "00:00:27.119", "25-
Sep-2006", "00:00:29.167", "25-Sep-2006", "00:00:31.215", "25-Sep-2006", "00:00:33.263", "25-Sep-2006", "00:07:12.621", "25-Sep-2006", "00:07:14.669",          ,          ,          ,          ,
,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,          ,
0,   1.8789e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   1.6582e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   1.4633e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   1.2914e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   1.1396e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   1.0057e+03,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   8.8754e+02,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   7.8326e+02,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   6.9122e+02,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
0,   6.1000e+02,   -9999,   -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999,      -9999
```

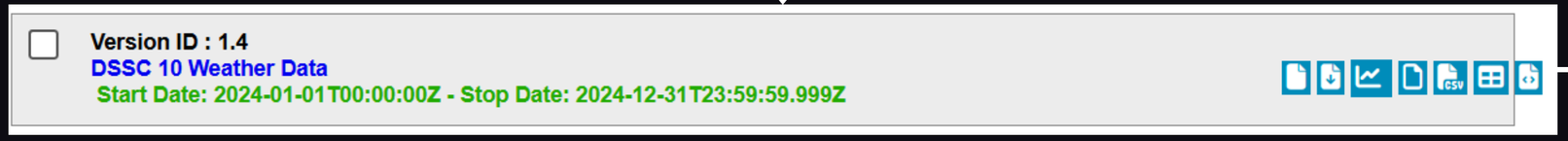
What you would like to know about Mars?

1. Was there life on Mars before, theoretically?
2. Is there a life on Mars?
3. What's the difference in physical level between Earth and Mars?
4. In how many million years Mars will be like Earth?
5. How long can people survive on Mars without a suitcase?

Date	Pres	T	T_err	Dust	Dust_err	H2Ovap	H2Ova	H2Oice	H2Oice_e	CO2ice	CO2ice_e	Alt	Lat	Lon
24.09.2006	1878,9	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	1658,2	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	1463,3	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	1291,4	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	1139,6	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	1005,7	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	887,54	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	783,26	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	691,22	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	610	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	538,32	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999
24.09.2006	475,07	197.415	0.202	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	0.232	-23.663	27.622
24.09.2006	419,25	196.990	0.180	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	1.502	-23.610	27.630
24.09.2006	369,98	196.090	0.175	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	2.767	-23.557	27.638
24.09.2006	326,51	194.772	0.174	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	4.027	-23.505	27.646
24.09.2006	288,14	193.115	0.177	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	5.279	-23.452	27.654
24.09.2006	254,29	191.216	0.184	-9999	-9999	-9999	-9999	-9999	-9999	-9999	-9999	6.520	-23.400	27.661
24.09.2006	224,41	189.175	0.195	1.3369e-03	8.3586e-0	-9999	-9999	1.1600e-0	5.7104e-0	-9999	-9999	7.748	-23.348	27.669
24.09.2006	198,04	187.117	0.209	1.2168e-03	5.0695e-0	-9999	-9999	1.4405e-0	4.1000e-0	-9999	-9999	8.964	-23.297	27.677
24.09.2006	174,77	185.118	0.226	1.1284e-03	4.2253e-0	-9999	-9999	1.6090e-0	3.5211e-0	-9999	-9999	10.168	-23.246	27.684
24.09.2006	154,23	183.235	0.247	1.0638e-03	4.0878e-0	-9999	-9999	1.8889e-0	4.0483e-0	-9999	-9999	11.361	-23.195	27.692
24.09.2006	136,11	181.520	0.269	1.0415e-03	2.5580e-0	-9999	-9999	2.2765e-0	3.7290e-0	-9999	-9999	12.543	-23.145	27.699
24.09.2006	120,12	180.025	0.294	9.4530e-04	1.9594e-0	-9999	-9999	2.3392e-0	3.2877e-0	-9999	-9999	13.715	-23.095	27.707
24.09.2006	106	178.758	0.319	7.9391e-04	1.8953e-0	-9999	-9999	1.8543e-0	3.1708e-0	-9999	-9999	14.878	-23.045	27.714
24.09.2006	93,547	177.741	0.293	6.5382e-04	1.7369e-0	-9999	-9999	1.0448e-0	2.5514e-0	-9999	-9999	16.035	-22.996	27.722
24.09.2006	82,555	176.982	0.226	5.3662e-04	1.1323e-0	-9999	-9999	4.8961e-0	1.4624e-0	-9999	-9999	17.188	-22.947	27.729

ALMA observation





DATE: 240101 DOY: 001 DSS 10

TIME (HHMM)	DEW PT (C)	TEMP (C)	PRESSURE (mb)	H20 PARTIAL PRES (mb)	RELATIVE HUM(%)
0000	2.5	9.1	905.4	7.4	63
0030	2.7	8.8	905.3	7.5	66
0100	3.1	8.2	905.4	7.7	70
0130	2.8	7.5	905.4	7.5	72
0200	2.6	7.1	905.6	7.4	73
0230	2.3	6.6	905.6	7.3	74
0300	2.1	6.1	905.5	7.1	75
0330	2.4	5.6	905.6	7.3	80
0400	2.8	5.6	905.4	7.5	82
0429	2.8	5.6	905.6	7.5	82
0459	3.1	5.6	906.0	7.7	84
0529	2.3	6.2	906.3	7.3	76
0559	1.5	5.2	906.2	6.9	77
0629	1.7	4.5	906.2	6.9	82
0659	2.0	4.1	906.0	7.1	86
0729	1.6	3.5	906.0	6.9	87
0759	2.5	4.9	905.8	7.3	84
0829	1.1	3.2	905.9	6.6	86
0859	2.1	3.7	905.8	7.1	89
0929	2.1	3.7	906.1	7.1	89
0959	2.2	3.6	906.1	7.2	90
1029	2.3	3.9	906.0	7.2	89
1059	1.1	3.6	906.1	6.6	83
1129	0.5	2.0	906.1	6.4	89
1159	1.4	2.2	905.9	6.8	94
1229	1.5	2.1	905.9	6.8	95
1259	1.4	1.9	906.1	6.8	96
1328	1.9	2.3	906.1	7.0	97
1358	1.1	1.5	906.3	6.6	97
1428	1.6	1.8	906.4	6.8	98
1458	2.2	2.2	906.7	7.2	100
1528	2.6	2.6	906.8	7.4	100
1558	3.5	3.5	907.0	7.9	100
1628	4.2	4.3	907.2	8.2	99
1658	4.3	5.3	907.6	8.3	93
1728	3.8	7.2	907.8	8.1	79
1758	2.9	8.5	907.8	7.6	68
1828	2.5	10.1	907.7	7.4	59
1858	1.4	10.9	907.4	6.8	52
1928	1.0	11.2	907.1	6.7	50
1957	0.8	11.4	906.6	6.6	48
2027	0.7	11.7	906.3	6.5	47

RESEARCH QUESTIONS:

1. Which organisms could have lived on the surface of Mars?

H0 - Bacterias and archaes would not survive on the surface of Mars
H1 - they will be able to live on the surface of Mars

2 Is there a difference between latitudes on Mars?

H0 - There is no significant difference between latitudes
H1 - At least 1 group is significantly different from others

- Mars is the popular topic in astronomy last years.
- Mars is the most similar planet to our Earth than other 6 planets in Solar System
- Humanity was not even considered about Moon and how people could be in Moon and in cosmos.

DATA CLEANING

	DAY	TIME HHMM	DEW PT C	TEMP C	PRESSURE mb	H20 PARTIAL PRES mb	RELATIVE HUM (%)	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: 10
0	2024-01-01	00:30:00	2.7	-11.2	905.3	7.5	63	NaN	NaN	NaN	NaN
1	2024-01-01	01:00:00	3.1	-11.8	905.4	7.7	66	NaN	NaN	NaN	NaN
2	2024-01-01	01:30:00	2.8	-12.5	905.4	7.5	70	NaN	NaN	NaN	NaN
3	2024-01-01	02:00:00	2.6	-12.9	905.6	7.4	72	NaN	NaN	NaN	NaN
4	2024-01-01	02:30:00	2.3	-13.4	905.6	7.3	73	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...
17398	2024-12-29	21:59:00	-6.4	-6.5	906.2	3.9	20	NaN	NaN	NaN	NaN
17399	2024-12-29	22:29:00	-6.6	-6.5	906.2	3.9	17	NaN	NaN	NaN	NaN
17400	2024-12-29	22:59:00	-6.9	-6.5	906.3	3.8	17	NaN	NaN	NaN	NaN
17401	2024-12-29	23:29:00	-7.6	-6.6	906.3	3.6	18	NaN	NaN	NaN	NaN
17402	2024-12-29	23:59:00	-7.6	-7.1	906.1	3.6	19	NaN	NaN	NaN	NaN

17403 rows × 11 columns

DATE	DEW PT C	TEMP C	PRESSURE mb	RELATIVE HUM (%)
2024-01-01	0.600000	-13.914894	907.226596	71.468085
2024-01-02	-0.410638	-12.510638	903.512766	60.319149
2024-01-03	-5.278723	-14.755319	903.782979	50.297872
2024-01-04	-5.527660	-14.700000	907.346809	50.617021
2024-01-05	-10.795745	-13.663830	906.825532	29.510638
...	...	...	...	...
2024-12-25	0.768750	-6.975000	900.650000	27.479167
2024-12-26	0.602083	-8.820833	904.956250	27.458333
2024-12-27	-1.166667	-8.747917	905.652083	24.708333
2024-12-28	1.210417	-8.868750	904.747917	20.270833
2024-12-29	-6.479167	-11.635417	905.477083	33.583333

364 rows × 4 columns

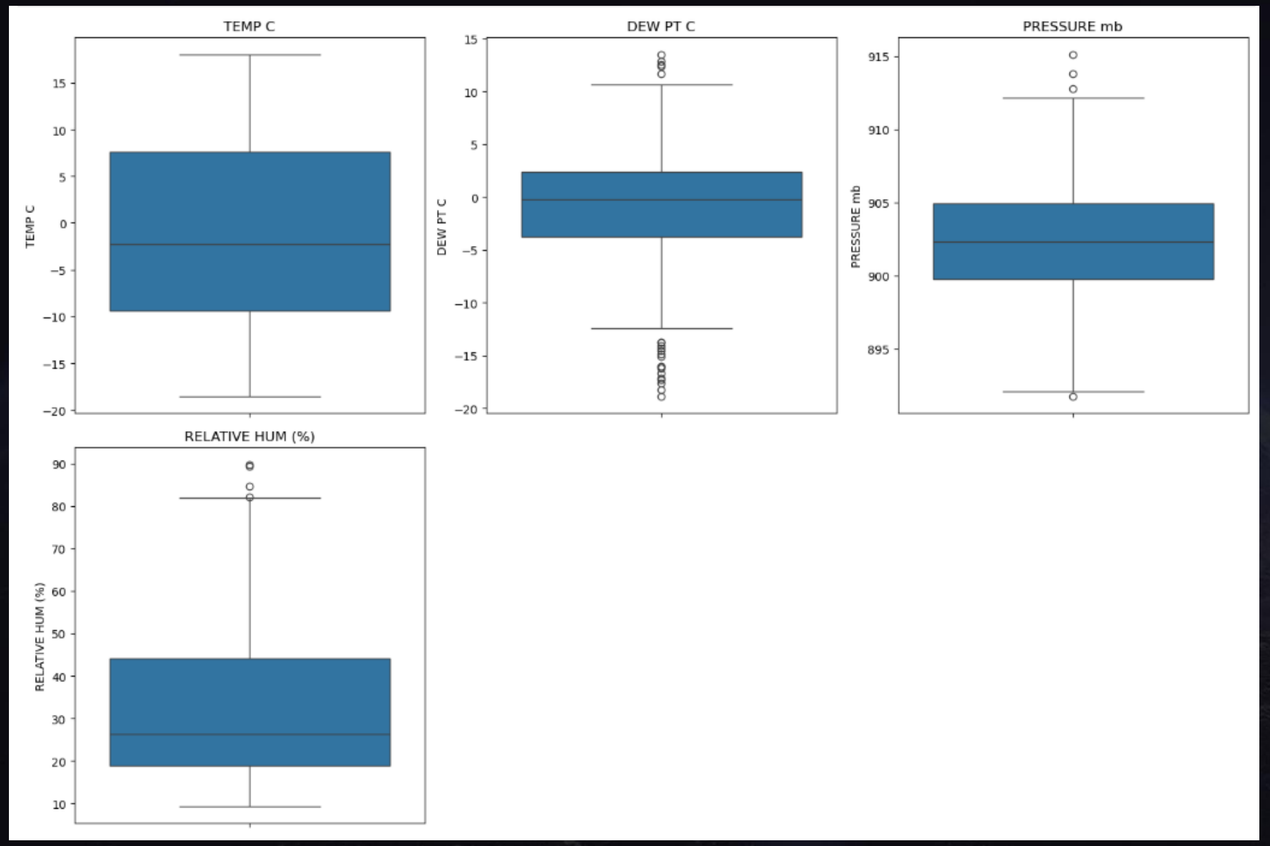
DEW PT C - Dew Point Temperature in Celsius. Dew point is a temperature where air becomes saturated with water vapor fully

TEMP C - just temperature in Celsius

PRESSURE mb - pressure column in millibar. 1 millibar is equal to 100 Pascal

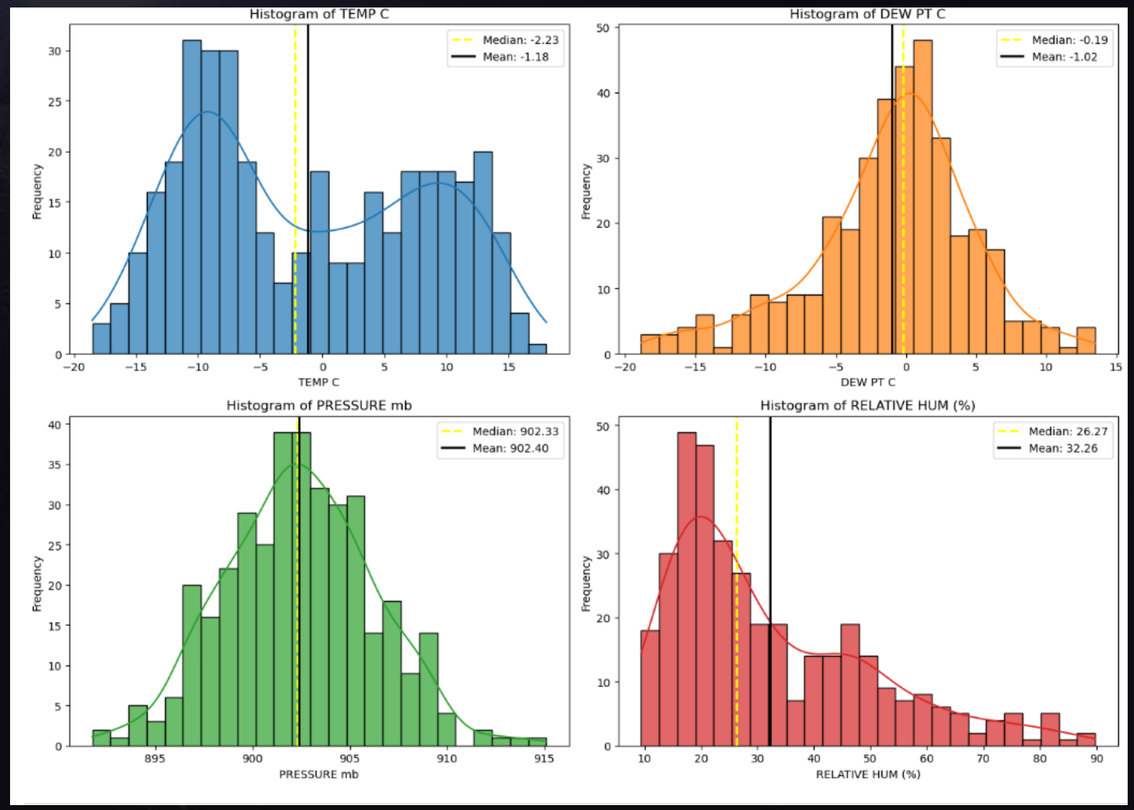
RELATIVE HUM (%) - describes the portion of the water vaper in the air.

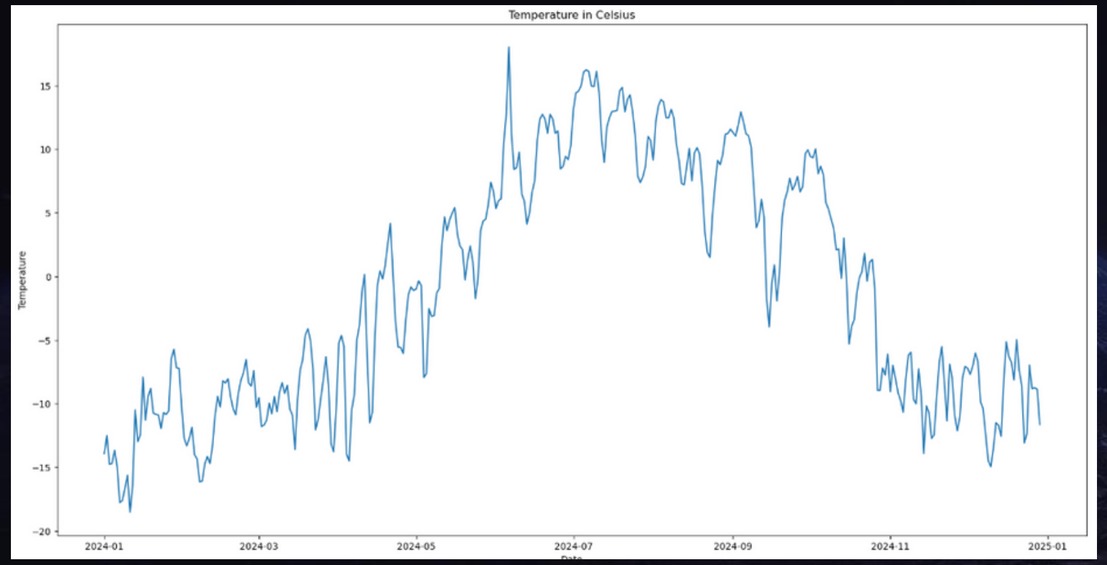
EDA

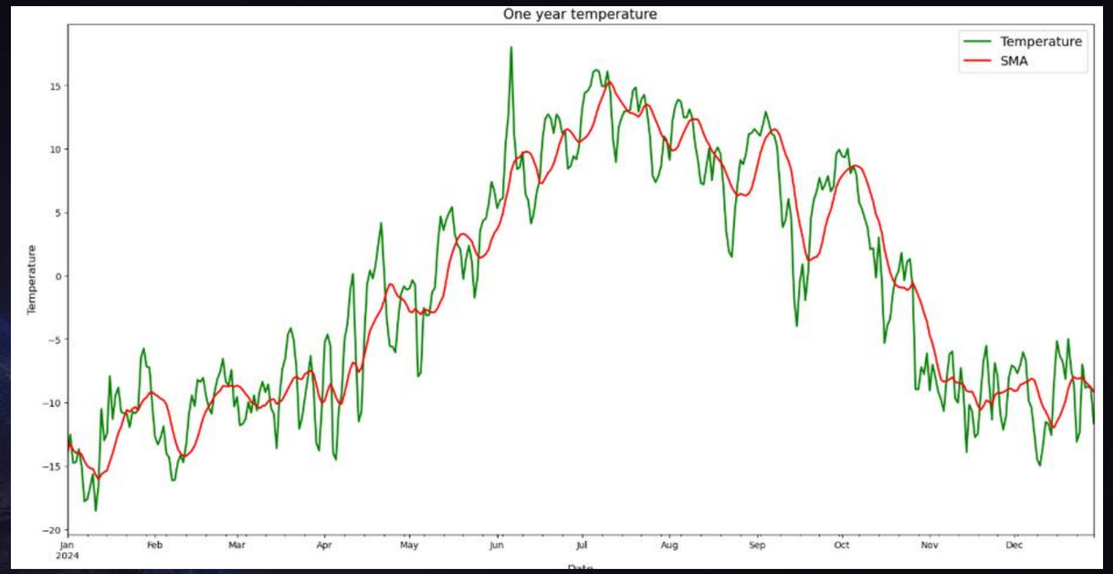


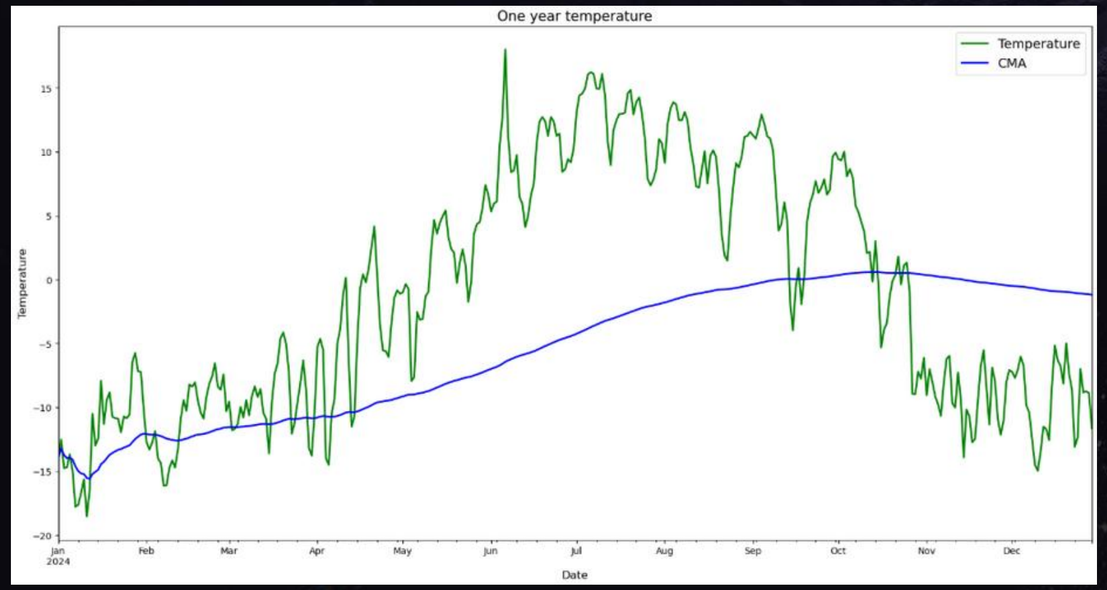
Boxplot: DEW PT C column has a lot of outputs that means every day the water vaper in the air may be very different.

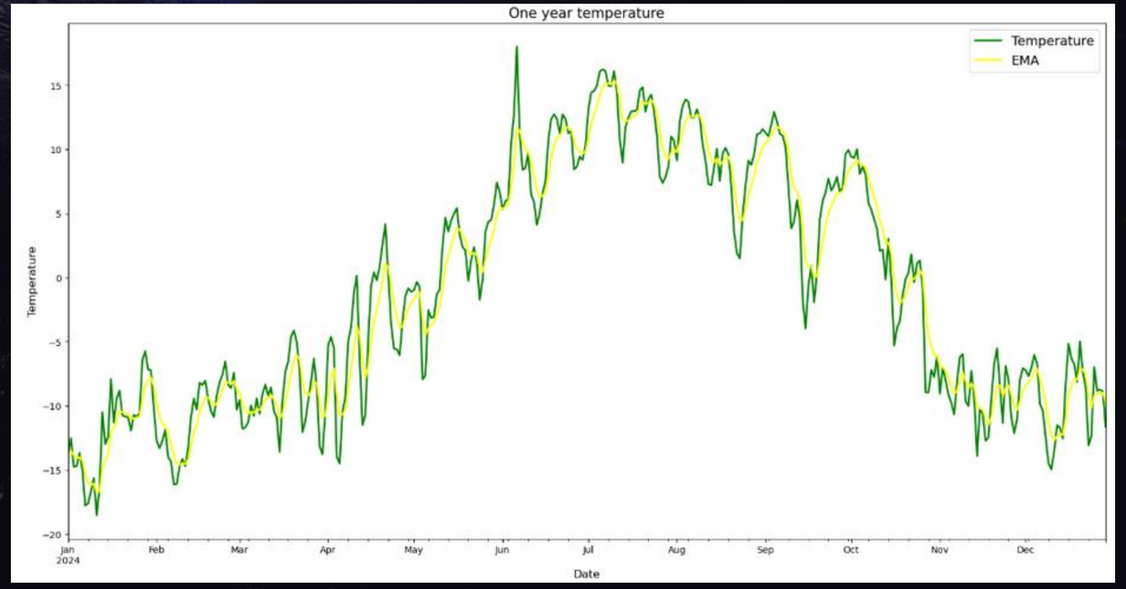
Histplot: temperature has -1,18 mean value and -2,23 median value




Temperature in Celsius
Temperature
15
10
5
0
−5
−10
−15
−20
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01

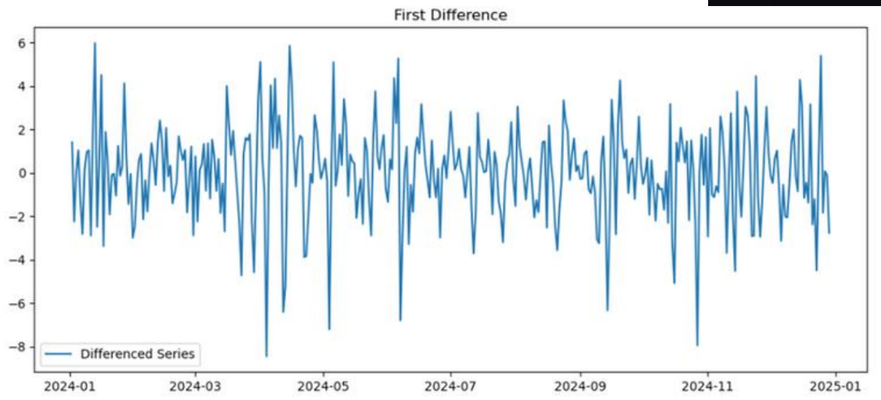

One year temperature
Temperature
SMA
Temperature
15
10
5
0
−5
−10
−15
−20
Jan
2024
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec


One year temperature
Temperature
CMA
Temperature
15
10
5
0
−5
−10
−15
−20
Jan
2024
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Date

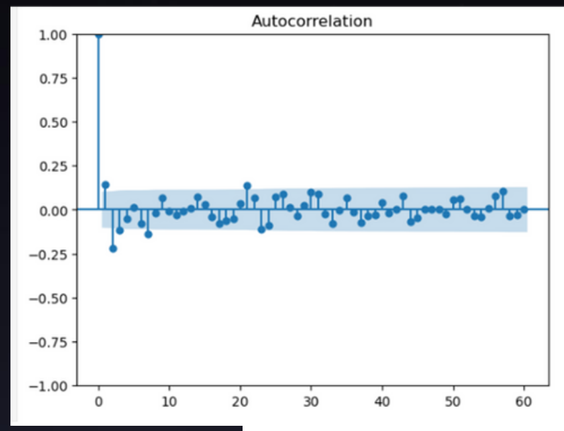

One year temperature
Temperature
EMA
Temperature
15
10
5
0
−5
−10
−15
−20
Jan
2024
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Date

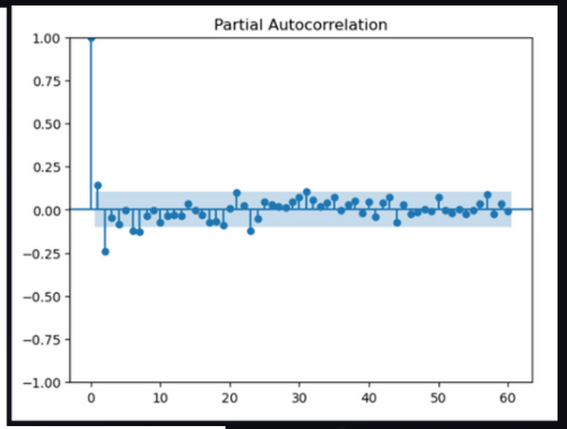
ARIMA

ADF Statistic: -1.675122302057953
p-value: 0.4439736341583351

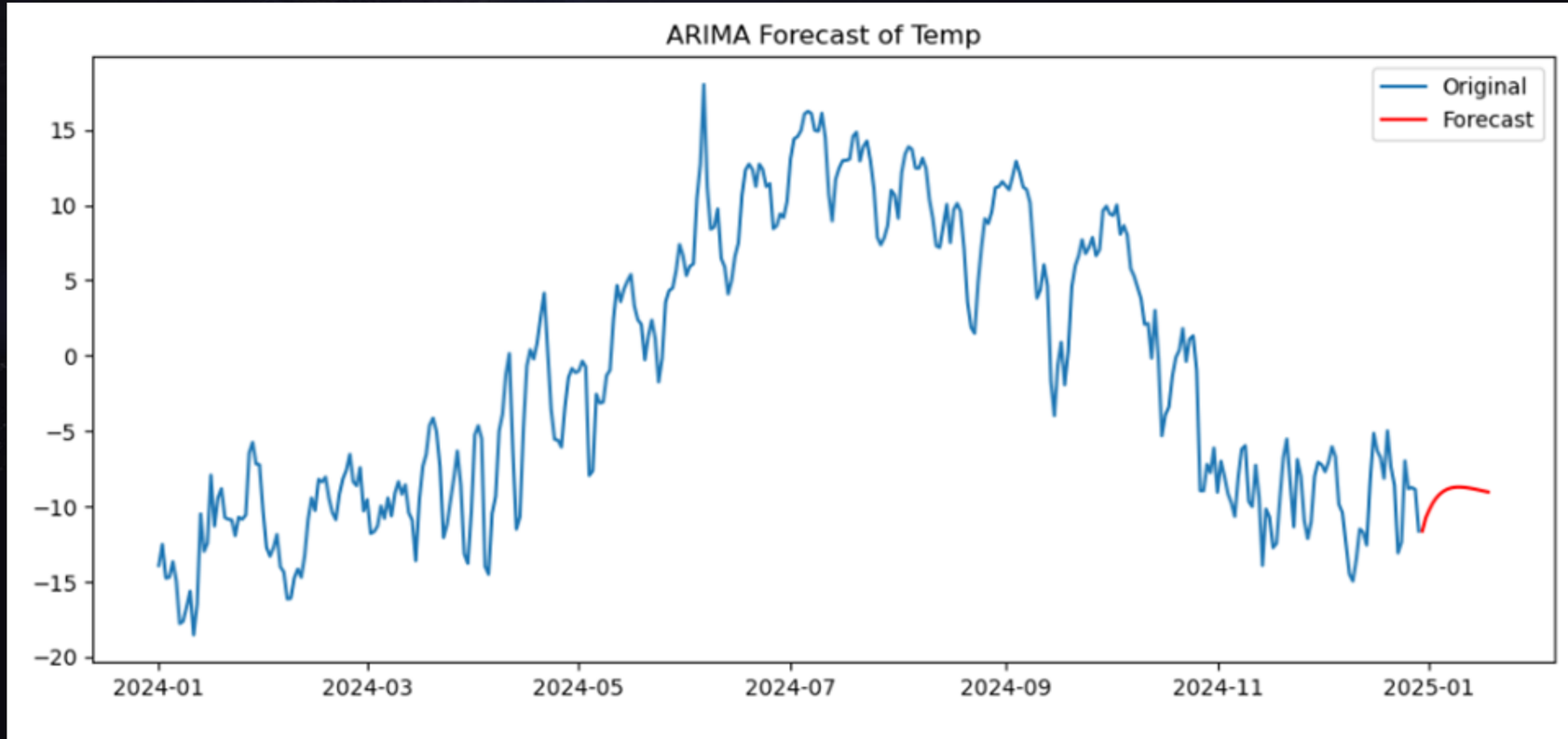


ADF Statistic after differencing: -9.847288947070277
p-value after differencing: 4.583655690812674e-17





R^2: 0.9533880548769508



	Date	Pres	T	T_err	Dust	Dust_err	H2Ovap	H2Ovap_err	H2Oice	H2Oice_err	CO2ice	CO2ice_err	Alt	l
0	2006-09-24	1878.900000	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	-9999.000	-9999.0
1	2006-09-24	1658.200000	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	-9999.000	-9999.0
2	2006-09-24	1463.300000	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	-9999.000	-9999.0
3	2006-09-24	1291.400000	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	-9999.000	-9999.0
4	2006-09-24	1139.600000	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	-9999.000	-9999.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
51445	2006-09-25	0.007002	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	86.413	-40.2
51446	2006-09-25	0.006179	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	87.218	-40.1
51447	2006-09-25	0.005453	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	88.024	-40.1
51448	2006-09-25	0.004812	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	88.830	-40.1
51449	2006-09-25	0.004247	-9999.0	-9999.0	-9999.0	-9999.0	-9999	-9999	-9999.0	-9999.0	-9999	-9999	89.636	-40.0

51450 rows × 15 columns

Kelvin to Celsius formula: C=K-273,15

	Date	Pres	Alt	Lat	Lon	Temp	Pres_kPa	T_score	P_score	Suitability
11	2006-09-24	475.07	0.232	-23.663	27.622	-75.735	475.07	0.106625	1.0	0.106625
12	2006-09-24	419.25	1.502	-23.610	27.630	-76.160	419.25	0.096000	1.0	0.096000
115	2006-09-24	538.32	0.251	-25.568	27.350	-67.011	538.32	0.324725	1.0	0.324725
116	2006-09-24	475.07	1.572	-25.513	27.358	-68.120	475.07	0.297000	1.0	0.297000
221	2006-09-24	475.07	0.425	-27.339	27.094	-74.075	475.07	0.148125	1.0	0.148125
...	...	...	...	...	...	...	...	...	...	...
51041	2006-09-25	475.07	0.614	-36.637	-138.035	-83.154	475.07	0.000000	1.0	0.000000
51042	2006-09-25	419.25	1.833	-36.586	-138.026	-83.587	419.25	0.000000	1.0	0.000000
51147	2006-09-25	419.25	1.039	-38.470	-138.335	-82.998	419.25	0.000000	1.0	0.000000
51252	2006-09-25	419.25	0.369	-40.325	-138.649	-86.496	419.25	0.000000	1.0	0.000000
51253	2006-09-25	369.98	1.567	-40.275	-138.638	-86.564	369.98	0.000000	1.0	0.000000

506 rows × 10 columns

for archaes:
T0 = -80 C
P0 = 200 Pa

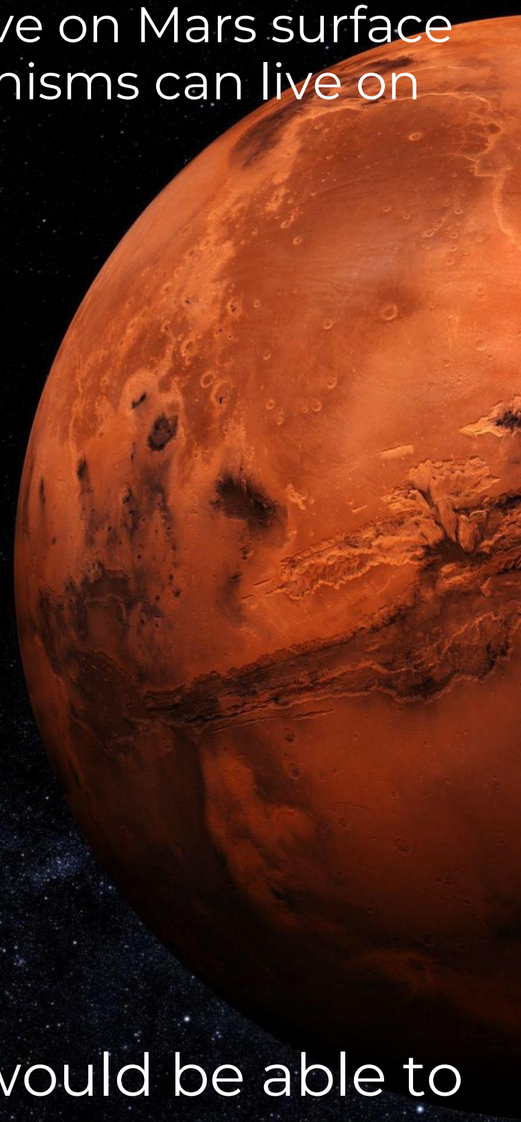
for people:
T0 = -25 C
P0 = 6.3 - 25 kPa

Hypothesis-test
H0: average suitability is less or equal to 0.2 and if it's less, bacterias could not live on Mars surface
H1: average suitability is higher than 0.2 and if it is yes, bacterias and microorganisms can live on Mars surface.

```
Shapiro p-value: 2.578051500472559e-16
```

Wilcoxon

```
p-value: 2.6320435321786388e-15
mean suitability: 0.3750971343873521
```

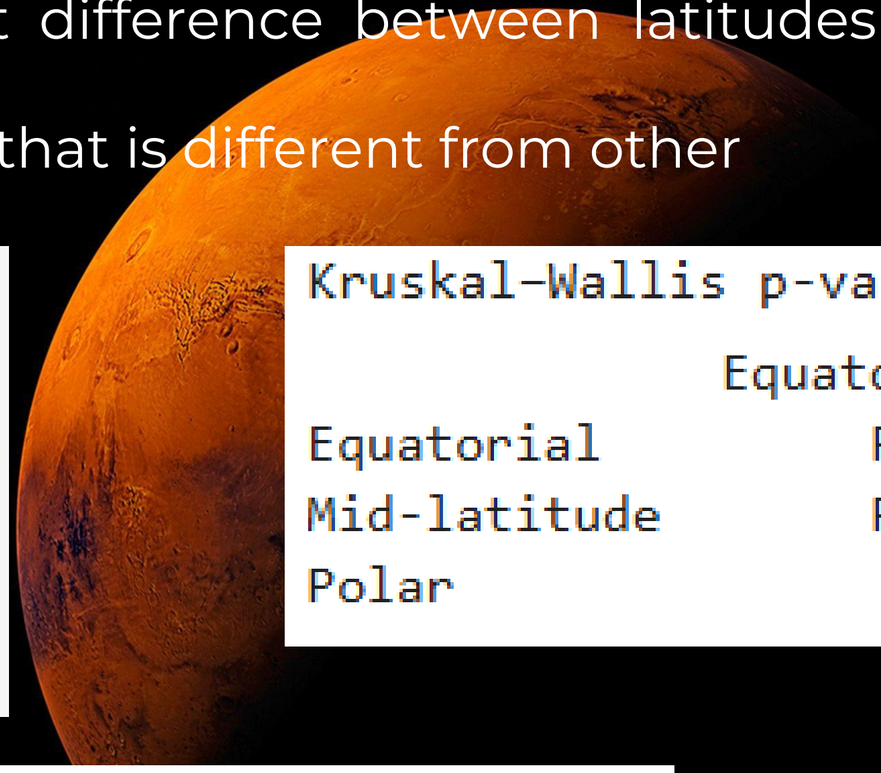


As p-value is less than 0.05, we reject Null-Hypothesis.
Conclusion: theoretically, bacterias, archaes and microorganisms would be able to live on Mars surface.

HYPOTHESIS TEST 2

H0: there is no significant difference between latitudes on Mars
H1: there is at least 1 group that is different from other

```python
def lat_zone(lat):
    if abs(lat) < 30:
        return "Equatorial"
    elif abs(lat) < 60:
        return "Mid-latitude"
    else:
        return "Polar"
```



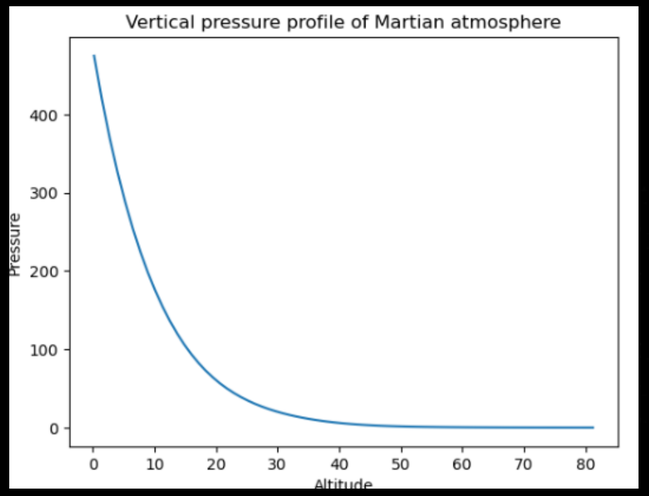
Kruskal-Wallis p-value: 0.004918376765383133

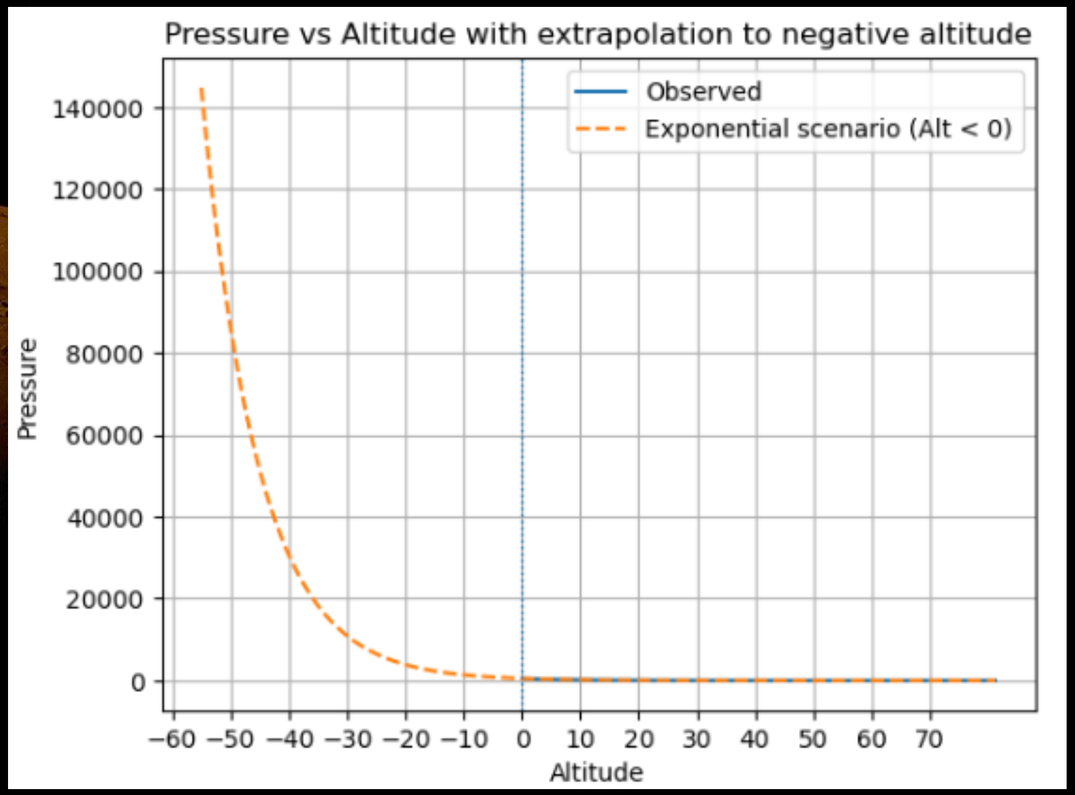
	Equatorial	Mid-latitude	Polar
Equatorial	False	False	True
Mid-latitude	False	False	True
Polar	True	True	False

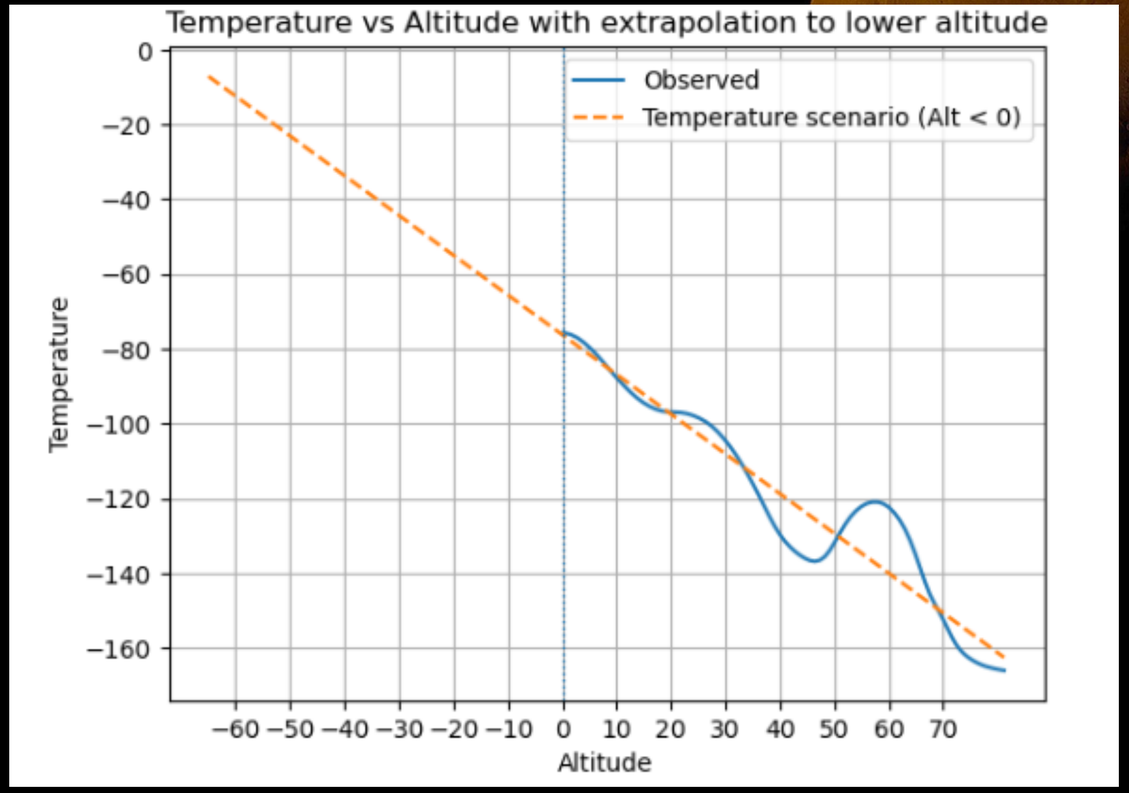
Shapiro Test: Statistic p-value =3.858545643040437e-07
❌ Data not normal (reject H_0)

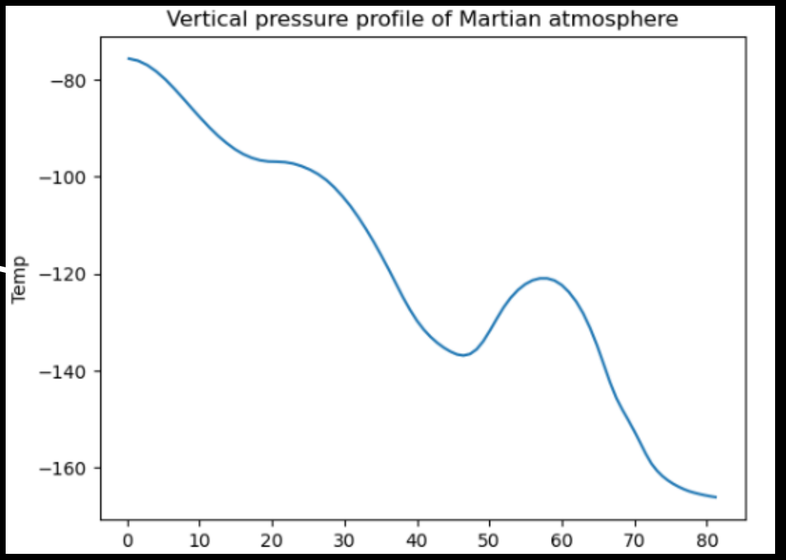
Shapiro Test: Statistic p-value =0.0015719895707228039
❌ Data not normal (reject H_0)

Shapiro Test: Statistic p-value =6.728213492555246e-16
❌ Data not normal (reject H_0)

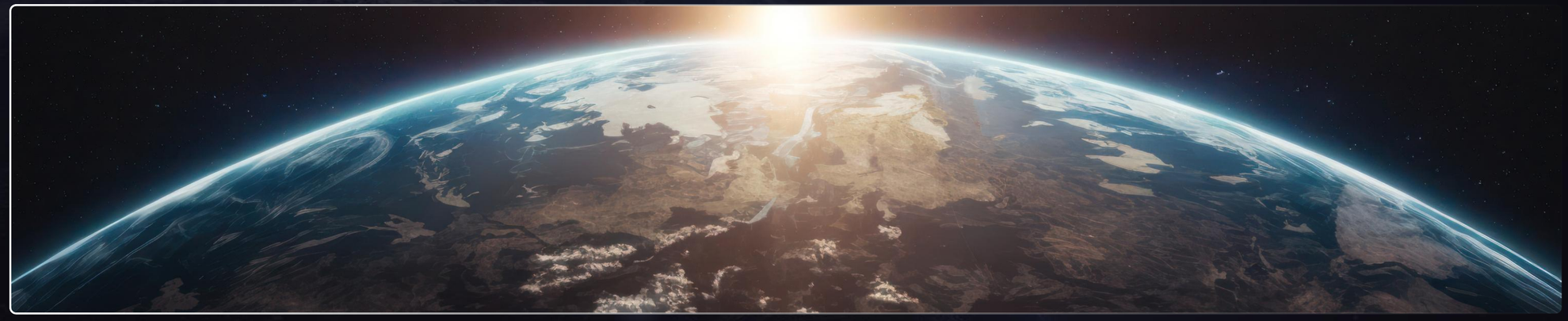

Vertical pressure profile of Martian atmosphere
Pressure
Altitude


Pressure vs Altitude with extrapolation to negative altitude
Observed
Exponential scenario (Alt < 0)
Pressure
Altitude


Temperature vs Altitude with extrapolation to lower altitude
Observed
Temperature scenario (Alt < 0)
Temperature
Altitude


Vertical pressure profile of Martian atmosphere
Temp

CONCLUSION



Conclusion: theoretically, humans would be able to live not in the surface of Mars, but below, under the surface of Mars, approximately in (-55) – (-60) meters.


THANK
YOU